

Lab 2 Prelab Assignment: Parameter Identification

1. Physical Source of Back EMF

Question: Explain physically the source of the back emf. Why is it proportional to the velocity?

A DC motor consists of a conductor (the rotor/armature) moving within a magnetic field produced by the stator. According to the principles of electromagnetism (Faraday's Law of Induction), a conductor moving through a magnetic field generates a voltage. Because the rotor is spinning, it "cuts" through the magnetic flux lines, inducing a voltage that opposes the main supply voltage.

The magnitude of this back emf (V_b) is proportional to the rate at which the magnetic flux is cut. Since the magnetic field is constant (in a permanent magnet motor), the rate of change is determined solely by how fast the rotor is spinning. This is expressed mathematically as:

$$V_b = K_b \cdot \omega$$

Where:

- K_b is the back emf constant.
- ω is the angular velocity.

2. System Connections

Question: List the connections that need to be made for each of the identification procedures based on Appendix C.

1. Time Domain Open-Loop Identification (DC Gain)

Objective: To determine the relationship between input voltage and steady-state angular velocity.

- **Input Connection:** Connect Point 2 (input potentiometer voltage source) to Point 19 (input to the power amplifier).
- **Output Monitoring:** Monitor motor speed by connecting the measuring device (oscilloscope or DAQ) to the Tachometer output at Points 24 and 25.
- **Procedure:** Manually vary the voltage from $-6V$ to $+6V$ using the potentiometer and record the resulting tachometer voltage response.

2. Time Domain Open-Loop Identification (Time Constant)

Objective: To observe the motor's step response and identify its rise time/time constant.

- **Input Connection:** Connect a function generator (providing a low-frequency square wave) between Point 19 and the system ground.

- **Output Monitoring:** Observe the Tachometer output (Points 24 and 25) on the oscilloscope.
- **Measurement:** Identify the time constant (τ) by measuring the time it takes for the signal to reach 63.2% of its steady-state value.

3. Frequency Domain Open-Loop Identification

Objective: To characterize the motor's bandwidth and break frequency.

- **Input Connection:** Connect the function generator (providing a 10V peak-to-peak sinusoid) between Point 19 and the system ground.
- **Output Monitoring:** Observe the gain and phase at the Tachometer output (Points 24 and 25).
- **Procedure:** Sweep the frequency from 0.2 Hz to 10 Hz and record the magnitude/phase relationship.

4. Time Domain Closed-Loop Identification

Objective: To identify parameters using second-order system characteristics (overshoot and peak time).

- **Feedback Connection:** Connect the output position pin (Point 5) to the summing op-amp block at Point 8. This establishes a unity negative feedback configuration.
- **Reference Input:** Apply a low-frequency square wave as the reference signal to Point 7.
- **Output Monitoring:** Measure the resulting shaft position at Point 5 to determine the percentage overshoot (P.O.) and peak time (T_p).

3. Second-Order System Analysis

Part (a): Parameter Comparison

The closed-loop transfer function under unity feedback is given as:

$$\Theta(s)/E_a(s) = K/(s^2 + \alpha s + K)$$

Comparing this to the standard second-order form:

$$\omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$$

- **Natural Frequency (ω_n):** By comparing the constant terms in the denominator, $\omega_n^2 = K$, therefore:

$$\omega_n = K^{1/2}$$

- **Damping Ratio (ζ):** By comparing the s coefficients, $2\zeta\omega_n = \alpha$, therefore:

$$\zeta = \alpha / 2\omega_n = \alpha / 2K^{1/2}$$

Part (b): Experimental Determination

To experimentally determine ζ and ω_n , you must measure the percentage overshoot (P.O.) and the peak time (T_p) of the step response.

- **Damping Ratio Calculation:**

$$\% \text{ Overshoot} = 100 \cdot e^{-\pi / (1-\zeta^2)}$$

- **Natural Frequency Calculation:**

$$T_p = \pi / \omega_n (1-\zeta^2)^{1/2}$$

Part (c): Computing Motor Parameters

Once the experimental values for ζ and ω_n are obtained, the system parameters are calculated as:

- $K = \omega_n^2$
- $\alpha = 2\zeta\omega_n$

4. MATLAB Simulations

Task: Plot the step responses for the following systems using the step function:

(a) First-order System:

$$H(s) = 1 / (s+3)$$

(b) Second-order System:

$$H(s) = 1 / (s^2 + 3s + 9)$$

